Product Specification

Company: Hangzhou Tuxi Information Technology Co., Ltd.
Product Name: 1MP Dual-Lens Facial Recognition Camera
Model Number: DS51M05D—V1.0
Version: V1.0

This product is a professional-grade 1MP dual-lens facial recognition camera board designed for industrial applications. It features a single-board structure for simplicity. The rated input voltage is DC 5V, and it can operate normally within DC 4.2V–5.5V. It connects via a standard USB interface to PCs, Android devices, etc., for master-slave half-duplex serial communication.

Product Features

1. Compliant with USB 2.0 standard and backward compatible with USB 1.1; conforms to UVC specifications
2. Wide input voltage range: DC 4.2V–5.5V
3. Customizable firmware for communication requirements
4. Low overall power consumption: 5V@240mA (including infrared LEDs)
5. Minimal USB bandwidth usage

Main Specifications

No.	Item	Specification Description
1	Focus	Manually adjustable
2	Imaging Speed	Less than 1.2 seconds
3	Color Resolution	Max: 1280(H) × 720(V) (1MP)Common: 1280×720, 640×480, 320×240, 160×120
4	B/W Resolution	Same as color resolution
5	Color Sensor	1/3.6” CMOS HD sensor GC1084
6	B/W Sensor	1/3.6” CMOS HD sensor GC1084
7	Color Pixel Size	3.0 × 3.0 µm
8	B/W Pixel Size	3.0 × 3.0 µm

No.	Item	Specification Description
9	Image Color Format	RGB 24-bit
10	Color Camera Dynamic Range	0–60 dB
11	B/W Camera Dynamic Range	0–50 dB
12	Image Format	YUY2 / MJPEG optional
13	Interface	USB 2.0
14	Light Source	6 high-power dot-matrix IR LEDs for uniform lighting
15	Supported OS	Windows XP, Win7 SP2, Vista, Win8, Linux 2.6.26, Android
16	Distortion Rate	< 15%
17	White Balance	Auto / Manual
18	Exposure Control	Auto / Manual
19	Signal-to-Noise Ratio	< 41 dB
20	Facial Recognition Distance	60 mm to 1.5 m (with standard lens)
21	Frame Rate	30 fps @ 1280×720 in MJPEG mode
22	Lens	Full-glass HD (available in IR or visible light versions)

Environmental Conditions

No.	Item	Technical Specification	Unit
1	Operating Temp.	-20 to +85	°C
2	Storage Temp.	-40 to +90	°C
3	Relative Humidity	15%–85%	%
4	Cooling Method	Natural air cooling	
5	Altitude	≤ 3000	m
6	MTBF	≥ 50,000	H

Electrical Characteristics

Input Characteristics

No.	Item	Specification	Unit
1.1	Rated Input Voltage	5 V	VDC
1.2	Input Voltage Range	4.2 V – 5.5 V	VDC

No.	Item	Specification	Unit
1.3	Maximum Current	500 mA	mA

Communication Protocol

Communication Requirements

- **Physical Layer:** USB 2.0
- **Cable:** 5-core shielded plug
- **Standard Length:** 2 meters
- **Recommended Length:** < 1 meter
- **Transfer Rate:** 480 Mbps
- **Data Mode:** Half-duplex communication
- **Protocol:** USB 2.0

Protocol Details

Communication follows USB 2.0 standards. The host (PC, Android, Linux, etc.) initiates all requests, while the camera (device) responds.
Basic process: Host sends request → Device responds.
The camera firmware includes device name, manufacturer name, version number, Vendor ID (VID), Product ID (PID), and a unique serial number ranging from 0x00000001 to 0x11111111 (user-definable and hardware-configured).

Operation Process:
Connect camera to USB port → camera powers on and initializes → host requests hardware ID and device info → if successful, open AMCAP → camera starts working.

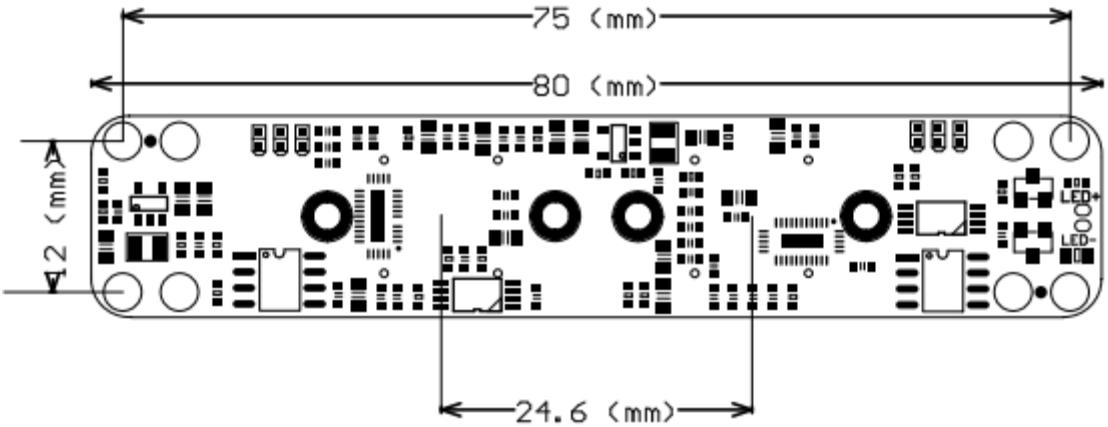
Mechanical Characteristics & Connector Definition

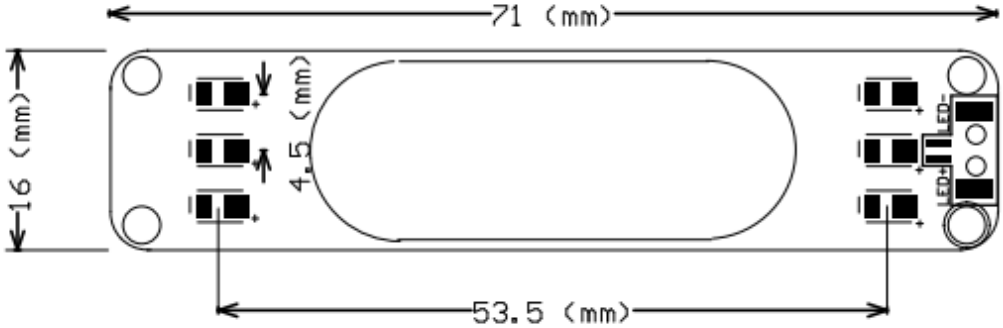
1. External Dimensions

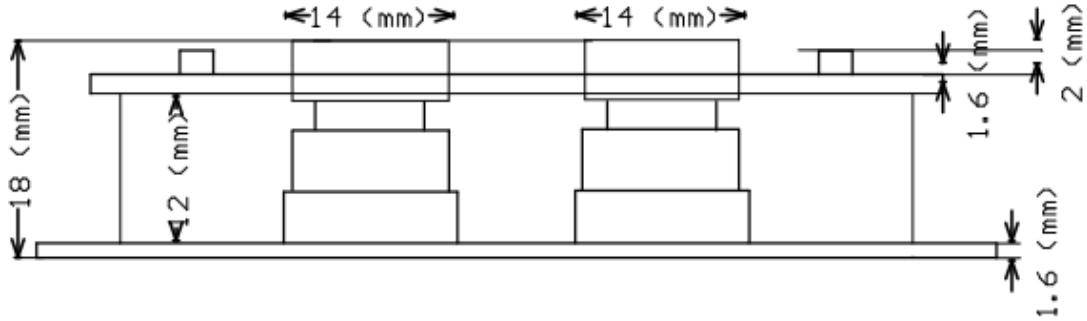
- **Maximum Dimensions:** 80 × 16 × 18 mm (L × W × H)
  (Customizable based on customer requirements)

- **Mounting Instructions:**
  Horizontal mounting; camera should face the target object.
  Adjust installation angle and height appropriately during setup.

2. Module Structure

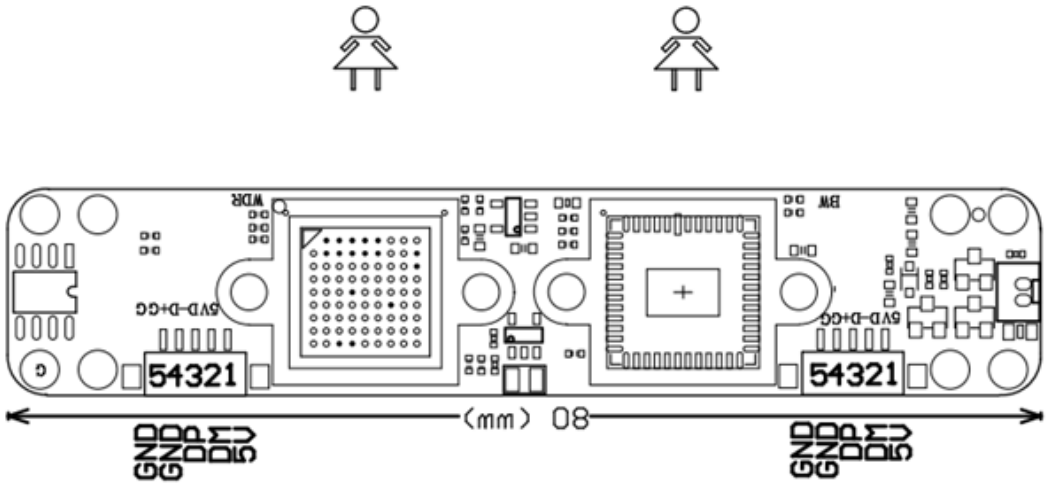






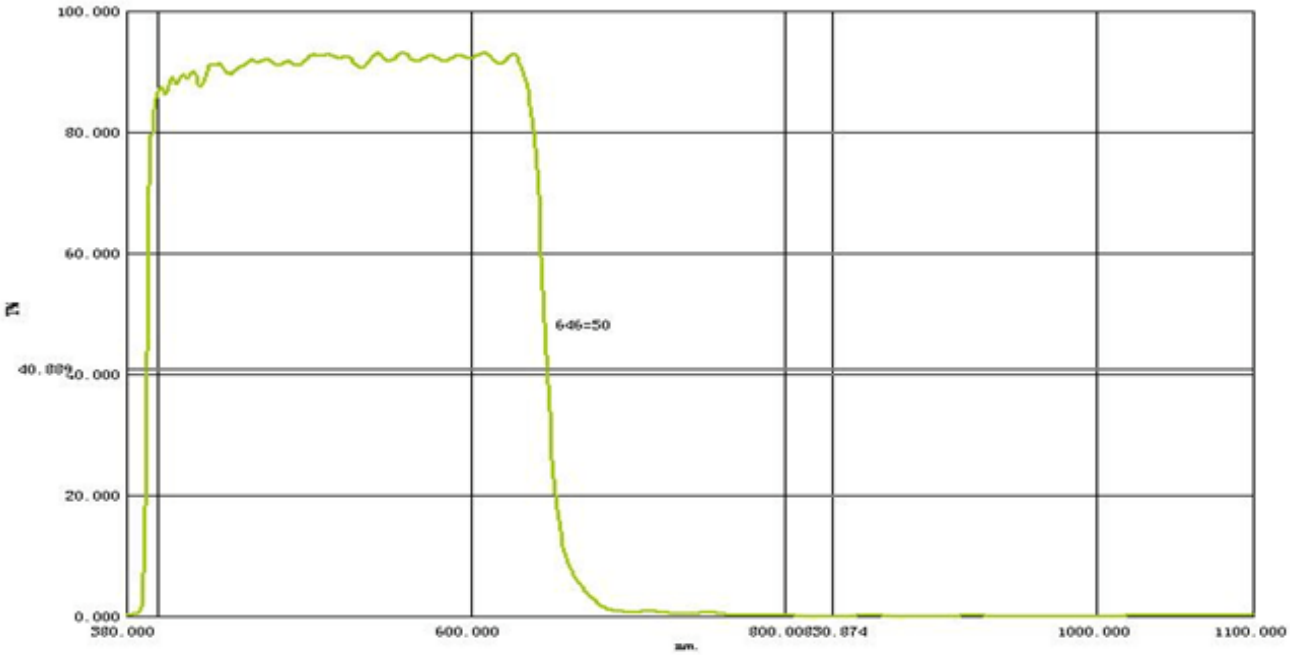
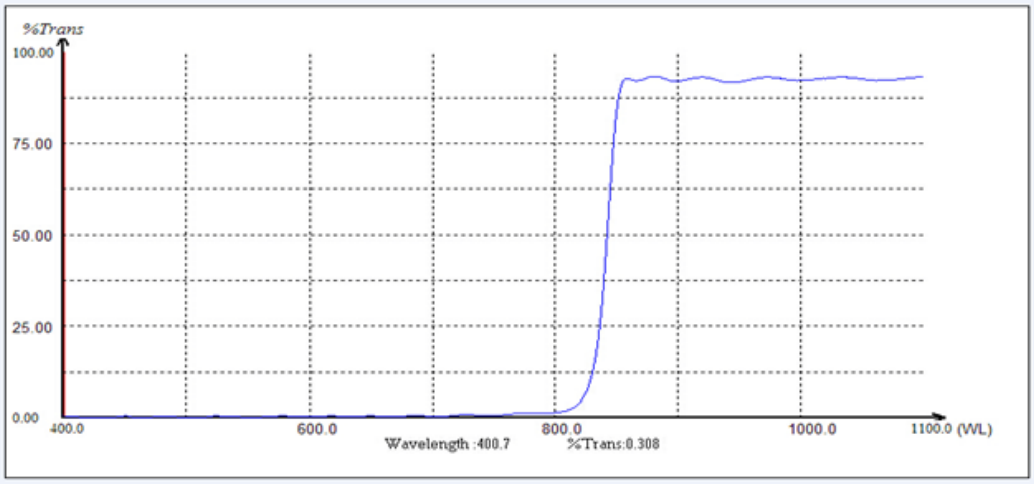
The final height may vary depending on the lens configuration and should be confirmed based on the actual product

3. Terminal Definitions (Connector Reference)



Optical Filter Spectrum & Lens Specifications

Color Filter Spectrum



Black & White (IR) Filter Spectrum